

Mathematics of Xgboost

	Exam (x)	Package (y)	$(\hat{y}_1)$ pred1 M ₁	$y - \hat{y}_1$ R ₁	$(\hat{y}_2)$ ML pred2	$y - \text{pred2}$ R ₂
①	6.7	4.5	7.3	-2.8	-2.05	6.685
②	9.0	11	7.3	3.7	3.7	8.41
③	7.5	6.0	7.3	-1.3	-2.05	6.685
④	5.0	8.0	7.3	0.7	0.7	7.51

Xgb Tree

Dataset

$$\text{Residuals} = \{-2.8, 3.7, -1.3, 0.7\}$$

$$\frac{\text{Similarity}}{\text{score}} = \frac{(\text{sum of residuals})^2}{\# \text{ residuals} + \lambda}$$

$\lambda \rightarrow 0$

$$\frac{\text{Similarity}}{\text{score}} \Rightarrow \frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4}$$

$$\Rightarrow \underline{0.02}$$

$$\left\{ \begin{array}{l} 6.7 \\ 9.0 \\ 7.5 \\ 5.0 \end{array} \right\} \xrightarrow[\text{data}]{\text{sort}} \left\{ \begin{array}{l} 5.0 \\ 6.7 \\ 7.5 \\ 9.0 \end{array} \right\}$$

$$\left. \begin{array}{l} 5.0 \\ 6.7 \end{array} \right\} \rightarrow 5.85$$

$$\left. \begin{array}{l} 6.7 \\ 7.5 \end{array} \right\} \rightarrow 7.1$$

$$\left. \begin{array}{l} 7.5 \\ 9.0 \end{array} \right\} \rightarrow 8.25$$

SS = 0.02

Exam score < 5.85

0.7 -2.8, 3.7

↓ ↓

$SS_L = \frac{(0.7)^2}{1}$ $SS_R = \frac{(-2.8)^2 + (3.7 - 1.3)^2}{3}$

$\Rightarrow 0.49$ $\Rightarrow 0.05$

Gain $\Rightarrow SS_L + SS_R - SS_{Root}$

$\Rightarrow 0.49 + 0.05 - 0.02$

$\Rightarrow 0.523$

SS = 0.02

Exam score < 7.1

-2.8, 0.7 3.7, -1.3

↓ ↓

$SS_L = 2.20$ $SS_R = 2.88$

Gain = $SS_L + SS_R - SS_{Root}$

$= 2.20 + 2.88 - 0.02$

$= 5.06$

SS = 0.02

Exam score < 8.25

-2.8, -1.3, 0.7 3.7

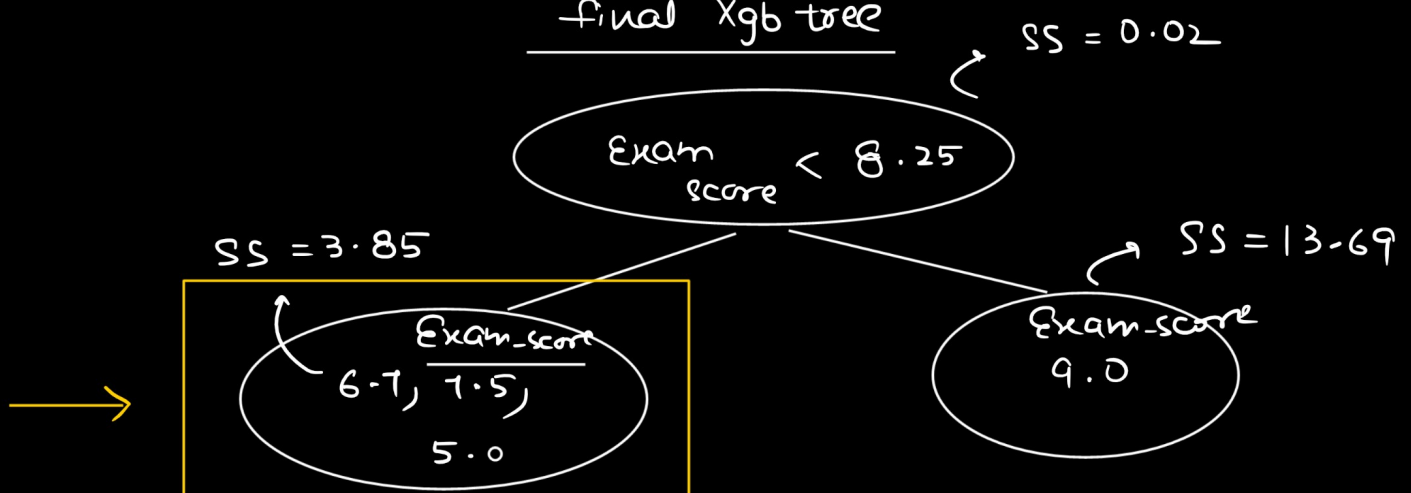
↓ ↓

$SS_L = 3.85$ $SS_R = 13.69$

Gain = $SS_L + SS_R - SS_{Root}$

$\Rightarrow 17.52$

final Xgb tree



5.0 } → 5.85

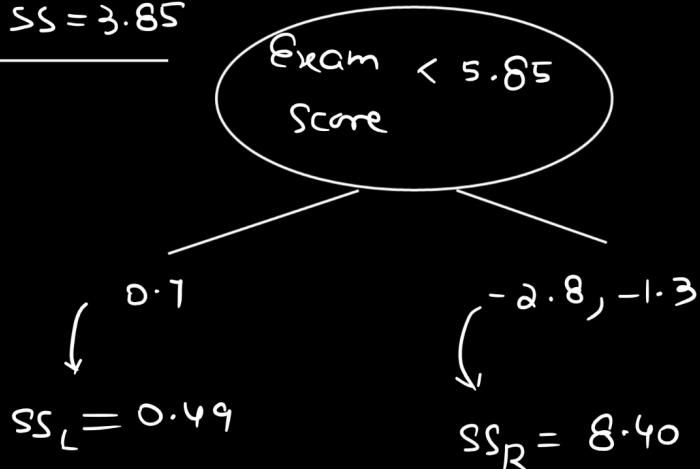
6.7 } → 7.1

7.5 }

C1

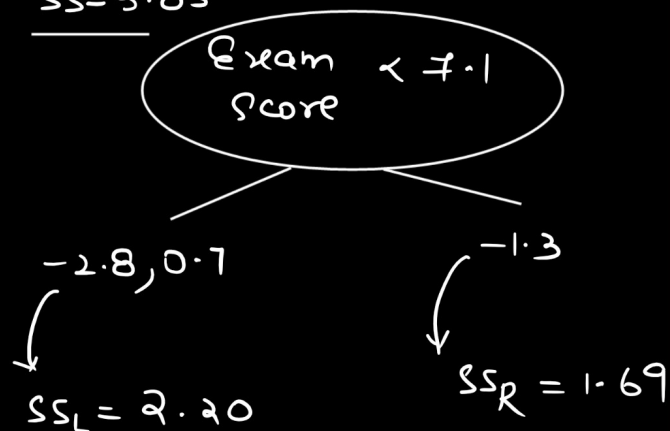
C2

$$SS = 3.85$$



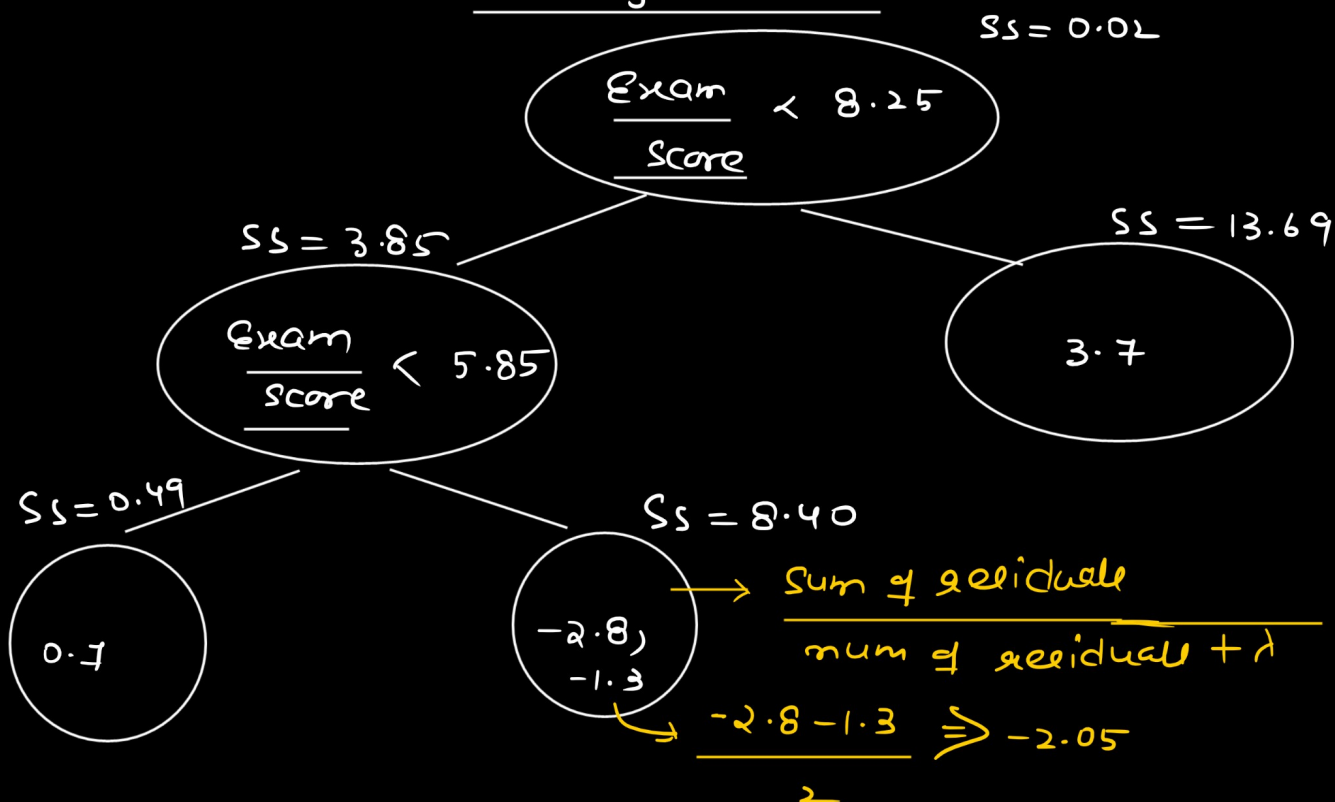
$$Gain = 0.49 + 8.40 - 3.85 \Rightarrow 5.04$$

$$SS = 3.85$$

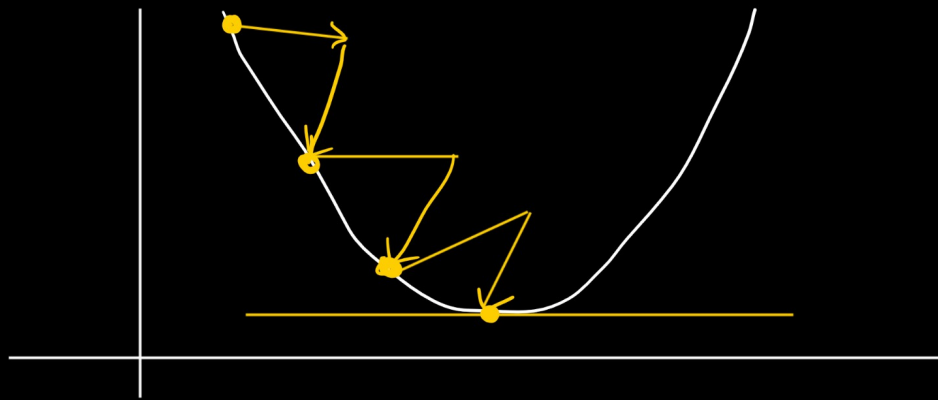


$$Gain = SS_L + SS_R - SS_{Root} \Rightarrow 0.045$$

final xgb tree



Optimizers (Residual, Loss-function, cost function)
→ Gradient Boosting



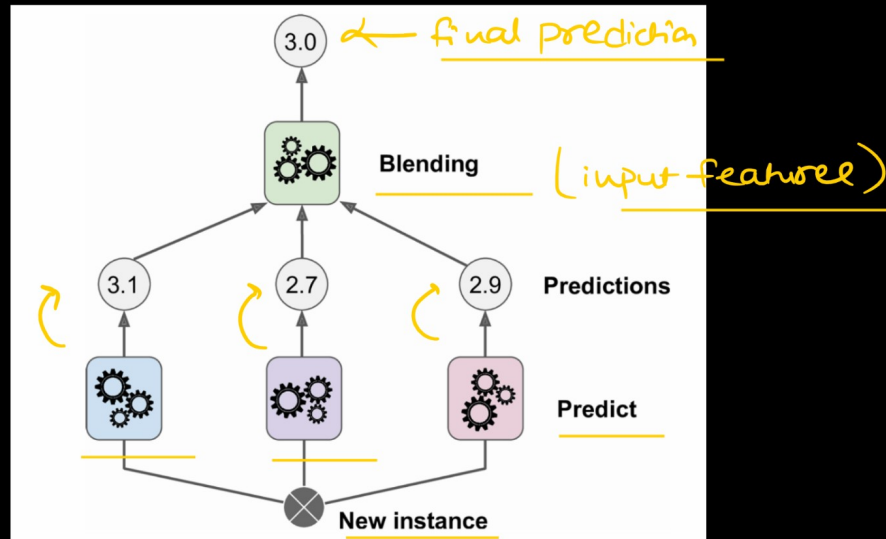
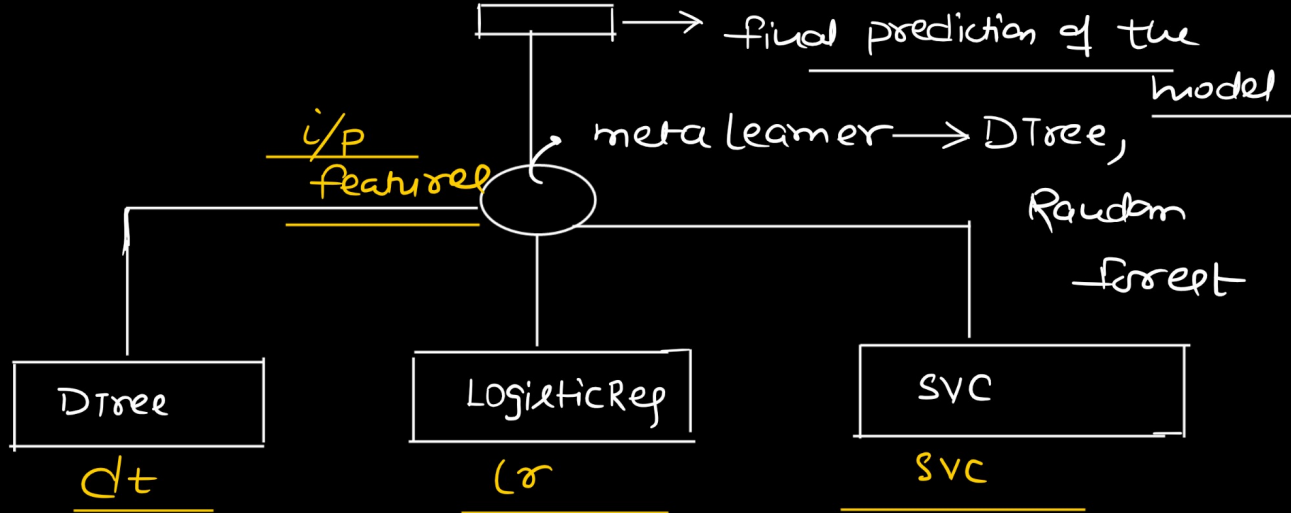
Hyperparameter tuning (gridsearch,

Random Forest randomized
searcher)

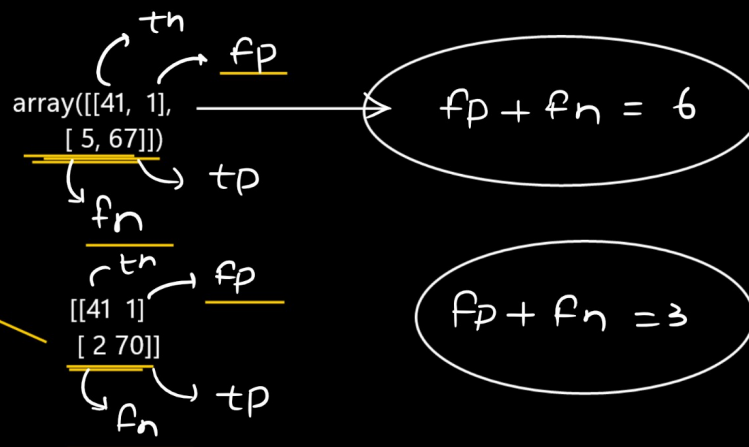
↳ $n_{\text{estimators}} = 100$

Xgboost

↳ $\lambda \geq 0, \eta$



tn, fp, fn, tp



$fp + fn = 3$ → Xgboost
perform
better

